

NSCBC Subsonic Outlet Boundary Conditions

OpenFOAM 10 — User and Theory Manual

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1 Overview

This library provides a partially reflecting (weakly non-reflecting) subsonic *outlet* for compressible reacting flow, implemented as three coupled boundary conditions — one per transported field that needs a characteristic treatment:

Type name	Applied to	Role
<code>pressureOutletNSCBC</code>	pressure p	acoustic relaxation to p_∞
<code>velocityOutletNSCBC</code>	velocity $\mathbf{U}$	acoustic + outgoing convection
<code>temperatureOutletNSCBC</code>	temperature T	entropy + acoustic transport

Species mass fractions Y_k at the outlet use OpenFOAM’s built-in **advective** condition (Section 4.2). The three conditions share the same parameters (`pInf`, `etaAc`, `lInf`, `fieldInf`) and must be applied together on the same patch.

Scope. The implementation is the subsonic-outlet case of the NSCBC framework of Poinso & Lele [1] with the Rudy–Strikwerda [3] pressure relaxation, evaluated with the variable-property thermodynamics of Baum, Poinso & Thévenin [2] under a **frozen-composition** assumption ($\partial Y_k / \partial n \approx 0$). Composition derivatives in the wave amplitudes (the full Baum et al. multi-component extension) are *not* included. Within that scope the formulation has been verified term-by-term against both papers; this manual documents that mapping.

2 Theory

2.1 Characteristic waves (Poinso & Lele)

On a boundary with outward normal n and normal velocity $u \equiv u_n$, the LODI (Local One-Dimensional Inviscid) analysis gives five characteristic waves with speeds

$$\lambda_1 = u - c, \quad \lambda_{2,3,4} = u, \quad \lambda_5 = u + c, \quad c^2 = \gamma p / \rho \quad (\text{P\&L Eq. 18}).$$

Their amplitude variations are (P&L Eqs. 19–23):

$$\mathcal{L}_1 = (u - c) \left(\frac{\partial p}{\partial n} - \rho c \frac{\partial u}{\partial n} \right), \quad \text{incoming acoustic} \quad (1)$$

$$\mathcal{L}_2 = u \left(c^2 \frac{\partial \rho}{\partial n} - \frac{\partial p}{\partial n} \right), \quad \text{entropy} \quad (2)$$

$$\mathcal{L}_5 = (u + c) \left(\frac{\partial p}{\partial n} + \rho c \frac{\partial u}{\partial n} \right), \quad \text{outgoing acoustic} \quad (3)$$

with $\mathcal{L}_3, \mathcal{L}_4 = u \partial u_{t_{1,2}} / \partial n$ the tangential-velocity waves (P&L Eqs. 21–22). All have units Pa/s.

Wave-numbering convention. This library uses the *Poinso & Lele* ordering, in which $\mathcal{L}_2$ is the *entropy* wave and $\mathcal{L}_3, \mathcal{L}_4$ are transverse. Baum et al. [2] number the equal-speed waves differently ($\mathcal{L}_2, \mathcal{L}_3$ transverse, $\mathcal{L}_4$ entropy, then N_s species waves). The code and this manual follow P&L throughout.

2.2 LODI relations and gradient relations

At a subsonic outlet $0 < u < c$, so $\lambda_1 < 0$ (incoming) and $\lambda_{2..5} > 0$ (outgoing). The outgoing amplitudes $\mathcal{L}_2, \mathcal{L}_5$ are computed from one-sided *interior* gradients; the incoming $\mathcal{L}_1$ must be supplied (Section 2.3). The boundary fields are then advanced with the primitive LODI relations (P&L Eqs. 25, 26, 29):

$$\frac{\partial p}{\partial t} + \frac{1}{2}(\mathcal{L}_1 + \mathcal{L}_5) = 0, \quad (4)$$

$$\frac{\partial u}{\partial t} + \frac{1}{2\rho c}(\mathcal{L}_5 - \mathcal{L}_1) = 0, \quad (5)$$

$$\frac{\partial T}{\partial t} + \frac{T}{\rho c^2} \left[-\mathcal{L}_2 + \frac{\gamma-1}{2}(\mathcal{L}_5 + \mathcal{L}_1) \right] = 0, \quad (6)$$

with transverse $\partial u_t / \partial t + \mathcal{L}_{3,4} = 0$ (Eqs. 27–28). The associated boundary-normal gradient relations (P&L Eqs. 34–36) are used to build the Neumann part of each condition:

$$\frac{\partial p}{\partial n} = \frac{1}{2} \left(\frac{\mathcal{L}_5}{u+c} + \frac{\mathcal{L}_1}{u-c} \right), \quad \frac{\partial u}{\partial n} = \frac{1}{2\rho c} \left(\frac{\mathcal{L}_5}{u+c} - \frac{\mathcal{L}_1}{u-c} \right), \quad \frac{\partial T}{\partial n} = \frac{T}{\rho c^2} \left[-\frac{\mathcal{L}_2}{u} + \frac{\gamma-1}{2} \left(\frac{\mathcal{L}_5}{u+c} + \frac{\mathcal{L}_1}{u-c} \right) \right]. \quad (7)$$

2.3 Pressure relaxation

$\mathcal{L}_1$ enters from outside and cannot come from interior data. Setting $\mathcal{L}_1 = 0$ (perfectly non-reflecting) leaves the mean pressure unconstrained and ill-posed; following Rudy & Strikwerda [3] it is relaxed weakly toward a far-field pressure p_∞ (P&L Eq. 40, identical to Baum et al. Eq. 26):

$$\mathcal{L}_1 = K(p - p_\infty), \quad K = \sigma(1 - \text{Ma}^2) \frac{c}{L}, \quad (8)$$

($\sigma = \text{etaAc}$, $L = \text{1Inf}$). Small K approaches the non-reflecting limit; large K approaches a fixed-pressure outlet. A residual reflection coefficient is therefore unavoidable — the deliberate trade-off between non-reflectivity and well-posedness.

Note (local vs. maximum Mach number). Poinso & Lele write Eq. (8) with the *maximum* Mach number of the flow; this library uses the *local* patch Mach number $\text{Ma} = u/c = \text{aP}/\text{cP}$, the now-common form. The two coincide for low-Mach outlets.

2.4 Variable thermodynamic properties (Baum et al.)

The only thermodynamic coefficients appearing in Eqs. (4)–(7) are γ , ρ , c (and R). Rather than treating them as constant (P&L), this library realises the advance of Baum et al. by evaluating them *face-by-face* from the live `fluidThermo` registry:

$$c = \sqrt{\gamma/\psi}, \quad \gamma = c^2\psi, \quad R = \frac{1}{\psi T},$$

where $\psi = \text{thermo:psi} = 1/(RT)$ is the compressibility. Because ψ and γ track local temperature and composition, c , the factor $(1 - \text{Ma}^2)$ in K , and the LODI scaling $T/(\rho c^2) = 1/(\gamma\rho R)$ are all spatially correct even across a flame front.

2.5 Reduction of Baum et al. to Poinso & Lele

Baum et al. [2] write the system in (ρ, T, u, v, w, Y_k) primitive variables with $N_s + 5$ waves; the temperature LODI is their Eq. 14,

$$\frac{\partial T}{\partial t} + \mathcal{L}_1 + \mathcal{L}_{N_s+5} + \sum_{i=4}^{N_s+3} \mathcal{L}_i = 0.$$

Under frozen composition $\partial Y_k/\partial n \rightarrow 0$: (i) the N_s species waves vanish; (ii) the species-gradient corrections to the acoustic and entropy waves vanish. Re-expressing the remaining (ρ, T) waves in the (p, u) variables of P&L through the fixed-composition ideal-gas relations $c^2 = \gamma RT$ and $dp = RT d\rho + \rho R dT$, Baum's Eq. 14 collapses term-by-term onto P&L Eq. 29 (6); the pressure and velocity equations likewise recover P&L Eqs. 25–26. **The frozen-composition outlet of Baum et al. is thus algebraically identical to the outlet of Poinot & Lele in P&L wave notation**; the physics of Baum et al. survives only through the variable properties of the previous subsection.

3 Implementation: LODI as an OpenFOAM mixed (Robin) condition

Each field is implemented as a `mixedFvPatchField`, whose face value is

$$\varphi_f = w \varphi_{\text{ref}} + (1 - w) [\varphi_C + \Delta (\partial \varphi / \partial n)_{\text{ref}}], \quad \Delta = \frac{1}{\text{deltaCoeffs}}, \quad \varphi_f - \varphi_C = \Delta \frac{\partial \varphi}{\partial n}, \quad (9)$$

with $w = \text{valueFraction}$, $\varphi_{\text{ref}} = \text{refValue}$, $(\partial \varphi / \partial n)_{\text{ref}} = \text{refGrad}$. The three coefficients are chosen so that (9) reproduces the discretised LODI relations. With $A \equiv K \Delta t / 2$ and $B \equiv \frac{u+c}{2} \Delta t / \Delta$ (Euler / Crank–Nicolson branch):

Pressure. $w = \frac{1 + A}{1 + A + B}$, $\varphi_{\text{ref}} = \frac{p^n + A p_\infty}{1 + A}$, $(\partial p / \partial n)_{\text{ref}} = -\rho c \frac{\partial u}{\partial n}$. Substituting into (9) and simplifying:

$$\frac{p_f - p^n}{\Delta t} + \underbrace{\frac{K}{2}(p - p_\infty)}_{\mathcal{L}_1/2} + \underbrace{\frac{u+c}{2} \left(\frac{\partial p}{\partial n} + \rho c \frac{\partial u}{\partial n} \right)}_{\mathcal{L}_5/2} = 0 \Rightarrow \frac{\partial p}{\partial t} + \frac{1}{2}(\mathcal{L}_1 + \mathcal{L}_5) = 0,$$

i.e. exactly P&L Eq. 25.

Normal velocity. $w = \frac{1}{1 + B}$, $\varphi_{\text{ref}} = u^n + \frac{K \Delta t}{2 \rho c} (p - p_\infty)$, $(\partial u / \partial n)_{\text{ref}} = -\frac{1}{\rho c} \frac{\partial p}{\partial n}$, which gives

$$\frac{\partial u}{\partial t} + \frac{1}{2 \rho c} (\mathcal{L}_5 - \mathcal{L}_1) = 0 \quad (\text{P\&L Eq. 26}).$$

The minus sign in each `refGrad` is *required*: it supplies the $\rho c \partial u / \partial n$ part of $\mathcal{L}_5$ only after combining with the $(\varphi_f - \varphi_C)$ term of (9). Read in isolation a `refGrad` resembles the $u - c$ impedance, but the assembled result is the correct outgoing-wave LODI.

Temperature. The patch advects at u ($w = 1/(1 + u \Delta t / \Delta)$) and carries the acoustic and entropy sources through `refValue/refGrad`; using $T/(\rho c^2) = 1/(\gamma \rho R)$ the assembly returns P&L Eq. 29 (6), and `refGrad` is the isentropic ($\mathcal{L}_2 = 0$) reduction of Eq. 36, $\partial T / \partial n = \frac{\gamma-1}{\gamma} \partial p / \partial n / (\rho R)$.

Transverse velocity. The NSCBC source terms are projected onto the patch-normal direction via `fieldInf`; tangential components reduce to zero-gradient extrapolation, consistent with $\partial u_t / \partial t + \mathcal{L}_{3,4} = 0$ in the limit of vanishing normal-transverse gradients.

Time schemes. Euler and CrankNicolson use first-order LODI time integration (weights above). `backward` uses BDF2 time levels ($1 \rightarrow \frac{3}{2}$ in w and the two-level φ^n, φ^{n-1} in `refValue`) with the same spatial operators. Other `ddt` schemes are rejected at run time.

4 Usage

Register the libraries in `system/controlDict`:

```
libs
(
    "libpressureOutletNSCBC.so"
    "libvelocityOutletNSCBC.so"
    "libtemperatureOutletNSCBC.so"
);
```

Apply all three on the outlet patch (plus advective for species):

```
// O/p
outlet { type pressureOutletNSCBC; pInf 101325; etaAc 0.25; lInf 0.05; fieldInf 1;
        value uniform 101325; }
// O/U
outlet { type velocityOutletNSCBC; pInf 101325; etaAc 0.25; lInf 0.05; fieldInf (1 0 0)
        ; value uniform (0 0 0); }
// O/T
outlet { type temperatureOutletNSCBC; pInf 101325; etaAc 0.25; lInf 0.05; fieldInf 1;
        value uniform 300; }
// O/Yi (each species)
outlet { type advective; phi phi; value uniform 0; }
```

4.1 Parameters

Parameter	Description	Required	Default
<code>pInf</code>	Target far-field pressure p_∞ [Pa]	yes	—
<code>etaAc</code>	Relaxation coefficient σ in K	yes	0.25
<code>lInf</code>	Relaxation length scale L [m] (e.g. flame-to-outlet distance)	yes	—
<code>fieldInf</code>	Direction multiplier (scalar 1; for $\mathbf{U}$, outward unit normal)	yes	—
<code>gamma</code>	Override γ ; if omitted, taken from <code>fluidThermo</code>	no	from thermo
<code>phi, rho, p, U</code>	Field-name overrides	no	<code>phi/rho/p/U</code>

4.2 Species boundary condition

The species LODI at the outlet is pure advection, $\partial Y_k / \partial t + u \partial Y_k / \partial n = 0$, which is exactly what OpenFOAM's `advective` condition solves — use it for every Y_k . A `zeroGradient` condition is the $\partial Y_k / \partial n \rightarrow 0$ limiting case and is acceptable when composition gradients at the outlet are truly negligible.

4.3 Build

```
cd pressureOutletNSCBC && wmake && cd ..
cd velocityOutletNSCBC && wmake && cd ..
cd temperatureOutletNSCBC && wmake && cd ..
```

Libraries are written to `$FOAM_USER_LIBBIN`.

4.4 Guidance

- Place the outlet where the flow is subsonic and composition gradients are small; the LODI approximation is exact only for boundary-normal waves, so oblique waves leave a small residual reflection.
- Increasing `etaAc` or decreasing `lInf` pulls the mean pressure to p_∞ more strongly but reflects more acoustic energy; $\sigma \approx 0.25$ (P&L §3.2) is a sound starting point.
- Keep `pInf` consistent with the value/internal field used to initialise p .

5 Validation (summary)

The formulation was assessed in one dimension: (i) a right-running Gaussian *acoustic* pulse and (ii) a convected Gaussian *entropy* pulse, both leaving the outlet with strongly reduced reflection relative to a baseline `fixedValue/zeroGradient` outlet (a weak residual consistent with the relaxation term); and (iii) a laminar premixed flame transported through the outlet, which crossed the boundary with its structure preserved and the pressure held near p_∞ . The condition therefore performs within the limits of the LODI model on which it is based.

References

- [1] T. J. Poinso and S. K. Lele, *Boundary conditions for direct simulations of compressible viscous flows*, J. Comput. Phys. **101**:104–129, 1992.
- [2] M. Baum, T. Poinso, and D. Thévenin, *Accurate boundary conditions for multicomponent reactive flows*, J. Comput. Phys. **116**:247–261, 1994.
- [3] D. H. Rudy and J. C. Strikwerda, *A nonreflecting outflow boundary condition for subsonic Navier–Stokes calculations*, J. Comput. Phys. **36**:55–70, 1980.